Fiscal Year:	FY 2009	**Task Last Updated:**	FY 07/10/2009
PI Name:	Johannigman, Jay A. M.D.		
Project Title:	Evaluation of Oxygen Concentrators at Altitude		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Smart Medical Systems and Technology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **ExMC**:Exploration Medical Capabilities		
Human Research Program Risks:	(1) **Medical Conditions**:Risk of Adverse Health Outcomes and Decrements in Performance Due to Medical Conditions that occur in Mission, as well as Long Term Health Outcomes Due to Mission Exposures		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Cincinnati	**State:**	OH
Zip Code:	45267-0558	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	04/01/2008	**End Date:**	03/31/2009
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-SMS00005		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	1) Original Aims The management of major illness and/or trauma in space, will likely result in the need for delivery of oxygen. On orbit delivery of oxygen relies on oxygen stores which also provide environmental oxygen. A major concern of oxygen delivery is the elevation of ambient oxygen concentration and increased risk of fire. Our previous work has found that in previously healthy individuals (similar to NASA astronauts) oxygen requirements can be met by oxygen at 3-4 liters per minute. This finding suggests that oxygen on-orbit could be provided by a oxygen concentrator. This device has the advantage of operating solely from electric power and since it concentrates oxygen from ambient air, does not result increases in environmental oxygen concentration reducing fire risk. The waste gas from a concentrator is nitrogen, resulting in a net environmental change of 0. Oxygen concentrators are used in home applications for patients with chronic lung disease and other respiratory disorders. We propose to evaluate the SeQual Eclipse and SeQual Integra oxygen concentrators at in an altitude chamber up to 30,000 feet. The investigators all have certifications for altitude chamber use and we will use the chamber at Brooks Air Force Base in San Antonio, TX. 2) Key Findings We evaluated currently on the shelf portable concentrators Results of the testing demonstrated the following important findings o The volume of oxygen produced per minute varied from 0.5 to 3.0 liters o As respiratory rate increases concentrator performance follows one of three characteristics o Oxygen concentration remains constant o Oxygen concentration remains constant, but only every other breath is rewarded with oxygen (the net effect being less oxygen to the patient) o Oxygen concentration falls precipitously with increasing breath rate (60-70%) o The SeQual Eclipse produces nearly three times the oxygen of second best performing device (Respironics EverGo) The delivered FIO2 from the concentrators was highest at sea level. FIO2 diminished as barometric pressure decreased. The Eclipse II failed to operate above 22000 ft. Power consumption was reduced at higher altitudes. At the highest flow settings, power consumption diminished by 30% during continuous flow and 31% during pulse dose in the Eclipse II and 19% in the Integra comparing sea level to 8,000ft. Battery duration on the Eclipse II at 8000 ft and 3 lpm was 1 hr 48 mins compared to 1 hr and 22 mins at sea level. Conclusion: Oxygen is a finite commodity, which is cumbersome and hazardous to transport. The relatively high FIO2 delivered by the POCs makes this method of O2 delivery a viable alternative to compressed O2 in select situations. However, POCs cannot deliver an FIO2 of 1.0, necessitating complementary compressed gas for these scenarios. At operational PB, POC function remains equivalent to operation at sea level. The PIO2 available to the patient however, remains constrained by lower PB as altitude increases. At sea level (PB of 750 mmHg) an FIO2 of 0.90 produces an alveolar O2 of 582 mmHg, at 8,000 ft (PB of 564 mmHg) an FIO2 of 0.90 produces an alveolar O2 of 415 mm Hg whereas, at an altitude of 32,000 ft (PB of 206 mmHg) an FIO2 of 0.90 produces an alveolar O2 of 93 mmHg. 3) Impact The findings suggest that even at extremes of altitude, oxygen concentrators are a safe alternative to compressed or liquid oxygen. At the extremes of mission altitudes, the device continues to operate with only a slight tail off in delivered oxygen concentration. An unsuspected finding was that at lower density the devices became more energy efficient owing to reduced resistance through the sieve beds, resulting in a longer battery life. Oxygen concentrators provide oxygen as long as there is sufficient electricity. In space, an oxygen concentrator is attractive as additional stores of oxygen are not required and use does not raise the oxygen concentration of the ambient environment. In austere environments (aircraft at altitude, remote areas, far forward military operations) oxygen concentrators provide a safe alternative to traditional oxygen sources.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Oxygen concentrators are capable of providing oxygen whenever electricity is available. Oxygen concentrators can be used instead of compressed oxygen in cylinders or liquid oxygen in a number of scenarios where transport of oxygen is hazardous or logistically challenging. This work supports the use of oxygen concentrators in far forward situations, in the back of aircraft, in extreme environments (climbing to altitude), and in emergency and mass casualty situations. This work has initiated new research into combining an oxygen concentrator with a ventilator for military and mass casualty operations. The use of concentrators in ground ambulances in the current conflict in the Middle East has been spurred by the success of this project.
Task Progress:	The project completed all proposed tasks. COTS concentrators will evaluated to identify the device capable of oxygen delivery of at least 3 L.min. An instrument to monitor oxygen concentration, flow, and volume at altitude was fabricated and tested. Two oxygen concentrators were tested at altitude. Results are to be presented at the ATACCC meeting in August and papers published in the Journal of Trauma in 2010.
Bibliography Type:	Description: (Last Updated: 07/10/2009)
Abstracts for Journals and Proceedings	Rodriquez D, Johannigman JA, Dorlac W, Branson RD. "Evaluation of oxygen concentrators at altitude." Advanced Technology for Combat Casualty Care Conference, St. Pete Beach, FL, August 10-12, 2009. Advanced Technology for Combat Casualty Care. In press, 2009. , Jun-2009

Papers from Meeting Proceedings	Rodriquez D, Johannigman JA, Dorlac W, Branson RD. "Evaluation of concentrators at altitude." Advanced Technology for Combat Casualty Care Conference, St. Pete Beach, FL, August 10-12, 2009. Advanced Technology for Combat Casualty Care Conference. In press, 2009. , Jun-2009
Papers from Meeting Proceedings	Rodriquez D, Johannigman JA, Dorlac W, Branson RD. "Maximizing oxygen delivery with a concentrator during mechanical ventilation." Advanced Technology for Combat Casualty Care Conference, St. Pete Beach, FL, August 10-12, 2009. Advanced Technology for Combat Casualty Care Conference. In press, 2009. , Jun-2009